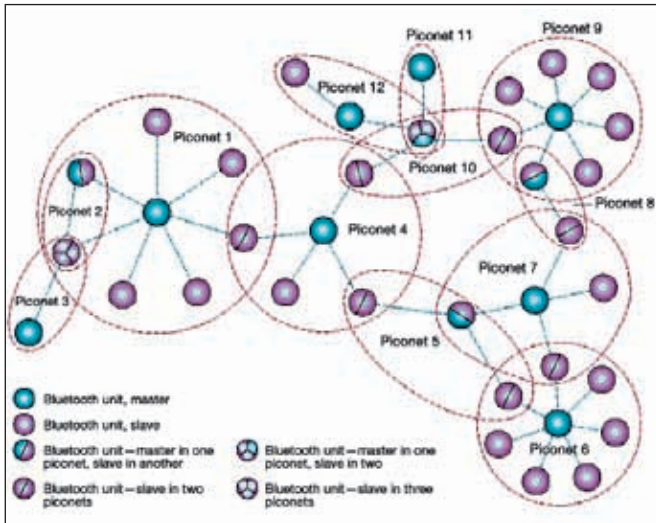


**Scatternet**

belong, simultaneously, to multiple piconets.

There are a few limitations with the piconet systems. Only three of them at a time can carry full-duplex voice transmissions. All the devices in a piconet share the same frequency-hop channel, which is established by the slaves synchronising their internal clocks to the master unit's clock. All piconets have a unique identity and thus multiple piconets can share the same physical space without interference.

To connect more than eight devices at one time, multiple piconets need to be created, and then all the master devices need to be connected together. Such a combination is called a scatternet.

Since Bluetooth uses 79 frequencies in total, a maximum of 10 piconets with 80 different Bluetooth devices can be included in any single scatternet. Beyond this number, saturation is reached.

However, multiple piconets that share the same physical space do not have to be connected to one another and can continue to operate independently without interference since each operates on a different frequency hop-channel.